MICHAEL F. PÉREZ

ML & Computer Vision Engineer Tampa, FL · Open to relocation
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Ph.D. in Computer Science with experience integrating computer vision into production software, building ML data and evaluation workflows, and researching video understanding and real-time ML systems.

TECHNICAL SKILLS

ML & computer vision: PyTorch, TensorFlow, OpenCV, scikit-learn, Hugging Face, FiftyOne, Label Studio
MLOps & systems: Kubernetes, MLOps, Airflow, MLflow, Docker, FastAPI, Flask, Linux, SLURM
Languages & data: Python, C++, SQL, JavaScript, NumPy, pandas, SciPy, Git

EXPERIENCE

Senior Engineer · Raft Aug 2026 – Nov 2026
Tampa, FL

- Developed and integrated computer vision data exploration into production software, including dataset visualization, clustering, filtering, natural-language querying, and geospatial analytics.
- Built ML data management and evaluation workflows for dataset curation, annotation review, model analysis, and debugging.
- Contributed to AI/ML development, testing, and integration, including object-detection experimentation and improvements to existing software systems.

Technologies used: Kubernetes, MLOps, Airflow, MLflow, Label Studio, FiftyOne.

Machine Learning Engineer Intern · CoVar May 2025 – Aug 2025
Durham, NC

- Applied infrared video super-resolution to long-range object classification; published with CoVar collaborators at the Military Sensing Symposium 2026.
- Trained lightweight RGB object-detection models for real-time drone detection targeting resource-constrained, on-device use.
- Built offline computer vision evaluation pipelines spanning preprocessing, inference, metrics, and efficiency analysis.

Graduate Research Assistant · University of Florida Aug 2020 – Aug 2026
Gainesville, FL

- Developed and evaluated video understanding models and scalable processing pipelines across the Ruiz HCI Lab, Data Science Research Lab, and Graphics, Imaging & Light Measurement Lab.
- Contributed to DARPA-funded CReLeRI and ENKIx systems, including UI and image/video analysis APIs, action segmentation, and real-time AR task guidance. Trained and tested ENKIx action-recognition models; the system reached 34% F1 in a multi-team evaluation. ENKIx is Ph.D. thesis work and a first-author manuscript in preparation.
- Built a video action-detection benchmark covering mAP, latency, throughput, memory, and FLOPs; first-author manuscript in preparation.

EDUCATION

Ph.D. in Computer Science · University of Florida Aug 2026
M.S. in Computer Science · University of Florida May 2023
B.S. in Computer Science · Florida Polytechnic University May 2020

SELECTED PUBLICATIONS

CReLeRI: Explainable, Concept-centric Representation, Learning, Reasoning, and Interaction Video Analysis System. First author · ACM Multimedia, 2025. doi:10.1145/3746027.3754479

Super-Resolution for Improving Vehicle Recognition at Large Stand-offs. With CoVar · Military Sensing Symposium, 2026.